

MILAD SOLEYMANI

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[Personal Website](#) - [GitHub](#) - [Linkedin](#) - [Google Scholar](#)

EDUCATION

- **M.Sc. Computer Science – Artificial Intelligence and Robotics** Sep 2024 – Present
Islamic Azad University, Tehran, Iran
◇ **GPA: 4.0/4.0**
- **B.Sc. Electrical Engineering** Sep 2018 – Sep 2023
K. N. Toosi University of Technology, Tehran, Iran
◇ **GPA: 3.14/4.0**

RESEARCH EXPERIENCE

- **Language of Reasoning in Large Language Models: Compliance, Code-Switching, and RL Training Dynamics** May 2026 – Present
Supervisor: Dr. Nava Eslami
 - ◇ Demonstrated that a model's architecture, not its prompt, decides whether it reasons in Persian: across 7 open 8B models, the one with a dedicated reasoning channel refused Persian chain-of-thought (4% compliance) while every inline-CoT model complied (68–85%)
 - ◇ Isolated the cause by ablating that channel on identical weights, raising compliance twenty-fold at no accuracy cost — evidence that a model can appear robust to non-English reasoning having never reasoned in it
 - ◇ Built the evaluation harness (vLLM and API inference, sentence-level language ID, paired significance tests); extending the result to training time with GRPO on Llama-3.1/3.2
- **Language-Aligned and Parameter-Efficient Representation Learning for Brain Signals** Oct 2025 – May 2026
Supervisor: Dr. Nava Eslami
 - ◇ Aligned an ATCNet brain-signal encoder with a frozen sentence-transformer text encoder (CLIP-style contrastive training), decoding motor tasks from natural-language descriptions rather than bare labels, toward zero-shot decoding of unseen commands
 - ◇ Engineered LoRA adapters for per-subject calibration (rank 8, 1.5% of parameters trainable), targeting the cross-subject variability that limits BCI deployment
- **Parameter-Efficient Fine-Tuning (PEFT) for Low-Resource Adaptation of Large Vision and Language Models** Jan 2025 – Present
Supervisor: Dr. Maliheh Sabeti, Dr. Nava Eslami
 - ◇ Adapted large pretrained vision models (Swin, ViT) to specialist medical tasks by training a small subset of parameters, cutting the compute needed to reach task performance
 - ◇ Designed an efficient Swin Transformer for cutaneous melanoma detection, submitted to *Biomedical Signal Processing and Control* (see Publications)

- Classification and Representation Learning for EEG** Mar 2020 – Jan 2025
Supervisor: Dr. Farhad Goodarzy, Dr. Andishe Partovi
 - ◇ Built preprocessing and training pipelines across multiple public EEG datasets for a transformer masked auto-encoder with vector quantization, learning task-agnostic embeddings
 - ◇ Benchmarked classification and feature-extraction methods, and implemented the time-frequency signal processing behind them
 - ◇ Contributed, as Data Scientist at KeyLead Health, to two EEG projects that resulted in peer-reviewed publications:
 - Partovi A., Burkitt A. N., Grayden D. **A Self-Supervised Task-Agnostic Embedding for EEG Signals**, IEEE/EMBS Conference on Neural Engineering (NER), 2023. [IEEE Xplore](#)
 - Partovi A., Grayden D. B., Burkitt A. N. **POC-CSP: A Novel Parameterised and Orthogonally-Constrained Neural Network Layer for Learning Common Spatial Patterns (CSP) in EEG Signals**, Journal of Neural Engineering, 2025. DOI: [10.1088/1741-2552/add8bc](#)
- Voice Recognition for Chronic Heart Failure (CHF) Phenotyping** Aug 2021 – May 2022
Supervisor: Dr. Andishe Partovi
 - ◇ Developed and evaluated models detecting chronic heart failure from voice recordings, with clinicians defining the phenotypes and validating the outputs

PUBLICATIONS

- Partovi A., Hosseini S. M., **Soleymani M.**, Liaghat K., Ziaee S., Habibi Panah Fard E., Sadeghpour Vajdi S., Goodarzy F. **A Deep Learning Algorithm for Classifying Grasp Motions using Multi-session EEG Recordings** 2021
 9th International Winter Conference on Brain-Computer Interface (BCI)
 DOI: [10.1109/BCI51272.2021.9385295](#)

Under review

- Soleymani M.**, Hoseyni K., Ziaee S., Eslami N. **Reasoning in Persian Is Optional: A Compliance Axis for Target-Language Chain-of-Thought in 8B Models** 2026
 Link: [Manuscript](#)
- Esmkhani M. R.**, **Soleymani M.**, Ziaee S., Eslami N., Boostani R., Sabeti M. **An Efficient Swin Transformer Architecture for Cutaneous Melanoma Detection** 2026
 Biomedical Signal Processing and Control (Elsevier), manuscript BSPC-D-26-10609
 Link: [Manuscript](#)

INDUSTRY EXPERIENCE

- AI Engineer** Oct 2025 – Present
QubitOrbit (Bozy AI), UAE-Dubai/Iran-Tehran
 - ◇ Built typed LLM agents with tool-grounded retrieval and clarification loops in place of unconstrained generation, parsing domain shorthand into structured records

- ◇ Led an event-driven market-intelligence pipeline — filtering, embedding, deduplication and online clustering in a vector database — with a multi-agent analyst running specialist models in parallel
- ◇ Designed the evaluation harness behind these systems, selecting prompts and models from measured accuracy and cost rather than intuition

- **AI Engineer** Sep 2024 – Sep 2025
MyFast Medical, Australia/Iran-Tehran
 - ◇ Fine-tuned large language models with reinforcement learning for empathic, patient-aligned generation in healthcare
 - ◇ Designed and deployed retrieval-augmented generation pipelines for medical domain tasks
- **Data Scientist (Computer Vision)** Nov 2023 – Feb 2024
ATHLX, Canada/Iran-Tehran
 - ◇ Modelled human body movement from video, building classifiers for motion-pattern interpretation
- **Data Scientist (Computer Vision)** Aug 2022 – Nov 2023
RUTILEA, Japan/Iran-Tehran
 - ◇ Built a generative pipeline (diffusion, VAE, ControlNet) for high-resolution industrial image synthesis
- **Data Scientist** Aug 2021 – Jul 2022
 KeyLead Health, Australia/Iran-Tehran
 - ◇ Built AI systems for biosignal and voice analysis (see Research: EEG decoding, chronic heart failure phenotyping)

TEACHING EXPERIENCE

- **Natural Language Processing** Sep 2025 – Present
 Islamic Azad University, Tehran, Iran — with Dr. Nava Eslami

GRADUATE COURSEWORK

• Deep Learning	20/20	• Deep Reinforcement Learning	19/20
• Large Language Models	20/20	• Evolutionary Computing	18.7/20
• Pattern Recognition	20/20	• Numerical Computation	18.5/20
• Digital Signal Processing	20/20	• Engineering Mathematics	18.4/20
• Machine Learning	19/20	• Advanced Programming	18.2/20

All courses graded 4.0/4.0 on the university's 4-point scale.

COURSES AND CERTIFICATES

- **Task-Oriented Course In Linux**, Quera Mar 2023
COURSE CERTIFICATE
- **DeepLearning.AI TensorFlow Developer Professional Certificate**, Coursera Jun 2021
COURSE CERTIFICATE

- **Advanced Python Programming and Object-Oriented Thinking**, Quera **Jun 2021**
COURSE CERTIFICATE
- **Deep Learning Specialization**, DeepLearning.AI, Coursera **Aug 2020**
COURSE CERTIFICATE
- **Machine Learning**, Coursera **May 2020**
COURSE CERTIFICATE

SKILLS

- **Programming Languages:**
 - ◇ Python, Java, C++
- **Machine Learning and Deep Learning:**
 - ◇ LLMs: Parameter-efficient fine-tuning (Adapters, LoRA, Prefix-Tuning), reinforcement learning from feedback (RLHF, GRPO), reasoning-model fine-tuning
 - ◇ Multimodal and Vision: Generative modelling (diffusion, VAEs, ControlNet), contrastive and self-supervised representation learning, vision transformers
 - ◇ EEG and Signal Processing: Time–frequency analysis, representation learning, transformer-based embeddings
- **LLM Systems and Retrieval:**
 - ◇ Agent design with tool calling and structured output, Model Context Protocol (MCP), prompt engineering, LLM evaluation and cost-accuracy benchmarking
 - ◇ End-to-end RAG: embedding models, chunking, hybrid embedding and BM25 retrieval; vector databases (Qdrant, FAISS, ChromaDB)
- **MLOps, Deployment and Infrastructure:**
 - ◇ Docker, Linux, Bash
 - ◇ Model monitoring, optimization, quantization, inference acceleration
- **Languages:**
 - ◇ English (TOEFL iBT: 93, Dec 16, 2023), Persian (Native)

HONORS AND AWARDS

- **Top Student, M.Sc. Artificial Intelligence and Robotics** **Sep 2024 – Present**
Islamic Azad University, Tehran, Iran
 - ◇ Named among the department’s top students, with a **GPA of 4.0/4.0**
- **2020 International BCI Competition — 6th Place** **Jan 2021**
 - ◇ Classifying hand grasping motion, i.e. cylindrical, spherical, lumbrical
 - ◇ Work published as *A Deep Learning Algorithm for Classifying Grasp Motions using Multi-session EEG Recordings*, 9th International Winter Conference on Brain-Computer Interface (see Publications)
- **Iran National University Entrance Exam (Konkur), B.Sc.** **2018**
 - ◇ Ranked top 0.3% (over 200,000 students)

REFERENCES

- **Dr. Behrooz Nasihatkon:** Assistant Professor in the Computer Science Department at K. N. Toosi University of Technology, CEO at Rahbin Sanat Nasir
Email: nasihatkon@kntu.ac.ir [linkedin](#)
- **Dr. Maliheh Sabeti:** Associate Professor in Computer Engineering at Islamic Azad University, North Tehran Branch
Email: sabeti@shirazu.ac.ir
- **Dr. Farhad Goodarzy:** Senior Researcher at the University of Melbourne, Australia; Senior AI Research Scientist at WEHI (Walter and Eliza Hall Institute of Medical Research)
Email: goodarzy@unimelb.edu.au [linkedin](#)
- **Dr. Andishe Partovi:** AI/ML Specialist at Google Cloud, Cofounder of Veris AI
Email: andipartovi@google.com [linkedin](#)
- **Dr. Nava Eslami:** Researcher at Sharif University of Technology's AI research center, Lecturer at Islamic Azad University
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